Summation Based On Binomial Series.

$$(1+x)^n = 1 + {}^nC_1 x + {}^nC_2 x^2 + \cdots + {}^nC_n x^n$$

$$= 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \cdots$$

$$(1-x)^{-n} = 1 + nx + \frac{n(n+1)}{2!}x^2 + \frac{n(n+1)(n+2)}{3!}x^3 + \cdots$$

$$(1-x)^{-p/q} = 1 + \frac{p}{1}(x/q) + \frac{p(p+q)}{1\cdot 2}(x/q)^2 + \cdots$$

$$(1-x)^{-1/2} = 1 + x/2 + \frac{1}{2}\cdot\frac{3}{4}x^2 + \frac{1\cdot 3\cdot 5}{2\cdot 4\cdot 6}x^3 + \cdots$$

Q. Sum the series $\frac{1}{2}\sin\alpha + \frac{1\cdot 3}{2\cdot 4}\sin 2\alpha + \frac{1\cdot 3\cdot 5}{2\cdot 4\cdot 6}\sin 3\alpha + \cdots$

$$C = 1 + \tfrac{1}{2}\cos\alpha + \frac{1\cdot 3}{2\cdot 4}\cos 2\alpha + \frac{1\cdot 3\cdot 5}{2\cdot 4\cdot 6}\sin 3\alpha + \cdots$$

$$S = \tfrac{1}{2}\sin\alpha + \frac{1\cdot 3}{2\cdot 4}\sin 2\alpha + \frac{1\cdot 3\cdot 5}{2\cdot 4\cdot 6}\sin 3\alpha + \cdots$$

$$C+iS = 1 + \tfrac{1}{2}(\cos\alpha + i\sin\alpha) + \frac{1\cdot 3}{2\cdot 4}(\cos 2\alpha + i\sin 2\alpha) + \cdots$$

$$= 1 + \tfrac{1}{2}e^{i\alpha} + \frac{1\cdot 3}{2\cdot 4}e^{i2\alpha} + \frac{1\cdot 3\cdot 5}{2\cdot 4\cdot 6}e^{i3\alpha} + \cdots$$

$$= (1-e^{i\alpha})^{-1/2} \qquad \cancel{(1-x)^{-1/2}}$$

$$= [1-(\cos\alpha + i\sin\alpha)]^{-1/2} \qquad \cancel{= 1 + \tfrac{1}{2}x + \frac{\frac{1}{2}\cdot\frac{3}{2}x^2}{2}} + \cancel{\tfrac{1}{2}\cdot\tfrac{3}{2}}\cdot\tfrac{7}{2}$$

$$= (1-\cos\alpha - i\sin\alpha)^{-1/2}$$

$$= (2\sin^2\alpha/2 - i\,2\sin\alpha/2\cos\alpha/2)^{-1/2} = [2\sin\alpha/2(\sin\alpha/2 - i\cos\alpha/2)]^{-1/2}$$

$$= \left(2\sin\alpha/2\right)^{-1/2}\left(\cos\left(\pi/2-\alpha/2\right)-i\sin\left(\pi/2-\alpha/2\right)\right]^{-1/2}$$

$$= \left(2\sin\alpha/2\right)^{-1/2}\left[\cos -\tfrac{1}{2}\left(\pi/2-\alpha/2\right)-i\sin -\tfrac{1}{2}\left(\pi/2-\alpha/2\right)\right]$$

$$= \left(2\sin\alpha/2\right)^{-1/2}\left[\cos\left(-\tfrac{\pi}{4}+\tfrac{\alpha}{4}\right)-i\sin\left(-\tfrac{\pi}{4}+\alpha/4\right)\right]$$

$$= \left(2\sin\alpha/2\right)^{-1/2}\left[\cos-\left(\pi/4-\alpha/4\right)-i\sin-\left(\pi/4-\alpha/4\right)\right]$$

$$C+iS = \left(2\sin\alpha/2\right)^{-1/2}\left[\cos\left(\pi/4-\alpha/4\right)+i\sin\left(\pi/4-\alpha/4\right)\right]$$

$$\Rightarrow S = \left(2\sin\alpha/2\right)^{-1/2}\cdot\sin\left(\pi/4-\alpha/4\right)$$

Q. Sum of series $\sin\alpha + \tfrac{1}{2}\sin 3\alpha + \frac{1\cdot 3}{2\cdot 4}\sin 5\alpha + \cdots$

$$S = \sin\alpha + \tfrac{1}{2}\sin 3\alpha + \cdots$$

$$C = \cos\alpha + \tfrac{1}{2}\cos 3\alpha + \frac{1\cdot 3}{2\cdot 4}\cos 5\alpha + \cdots$$

$$C+iS = (\cos\alpha + i\sin\alpha) + \tfrac{1}{2}(\cos 3\alpha + i\sin 3\alpha) + \frac{1\cdot 3}{2\cdot 4}(\cos 5\alpha + i\sin 5\alpha) + \cdots$$

$$= e^{i\alpha} + \frac{1}{2}e^{i3\alpha} + \frac{1\cdot 3}{2\cdot 4}e^{i5\alpha} + \cdots$$

$$= e^{i\alpha}\left(1 + \tfrac{1}{2}e^{i2\alpha} + \frac{1\cdot 3}{2\cdot 4}e^{i4\alpha} + \cdots\right)$$

$$= e^{i\alpha}\left[1 + \tfrac{1}{2}\left(e^{i\alpha}\right)^2 + \frac{1\cdot 3}{2\cdot 4}\left(\left(e^{i\alpha}\right)^2\right)^2 + \cdots\right]$$

$$= e^{i\alpha}\left(1-e^{i2\alpha}\right)^{-1/2}$$

$$= e^{i\alpha}\left(1-\cos 2\alpha - i\sin 2\alpha\right)^{-1/2}$$

$$= e^{i\alpha}\left[2\sin^2\alpha - i.2\sin\alpha\cos\alpha\right]^{-1/2}$$

$$= e^{i\alpha}(2\sin\alpha)^{-1/2}\left(\sin\alpha - i\cos\alpha\right)^{-1/2}$$

$$= e^{i\alpha}(2\sin\alpha)^{-1/2}\left(\cos(\pi/2-\alpha) - i\sin(\pi/2-\alpha)\right)^{-1/2}$$

$$= e^{i\alpha}(2\sin\alpha)^{-1/2}\left(\cos -\tfrac{1}{2}(\pi/2-\alpha) - i\sin -\tfrac{1}{2}(\pi/2-\alpha)\right]$$

$$= e^{i\alpha}(2\sin\alpha)^{-1/2}\left[\cos(-\pi/4+\alpha/2) - i\sin(-\pi/4+\alpha/2)\right]$$

$$= \left(\overset{e^{i\alpha}}{\cancel{\cos\alpha + i\sin\alpha}}\right)(2\sin\alpha)^{-1/2}\left[\cos(\pi/4-\alpha/2) + i\sin(\pi/4-\alpha/2)\right]$$

$$= e^{i\alpha}(2\sin\alpha)^{-1/2} \cdot e^{i\cdot(\pi/4-\alpha/2)}$$

$$= (2\sin\alpha)^{-1/2}\, e^{i(\pi/4-\alpha/2+\alpha)}$$

$$= (2\sin\alpha)^{-1/2}\, e^{i(\pi/4+\alpha/2)}$$

$$= (2\sin\alpha)^{-1/2}\left[\cos(\pi/4+\alpha/2) + i\sin(\pi/4+\alpha/2)\right]$$

$$\Rightarrow S = (2\sin\alpha)^{-1/2}\sin(\pi/4+\alpha/2)$$

Q. $\sin\alpha + n\sin(\alpha+\beta) + \dfrac{n(n-1)}{1\times 2}\sin(\alpha+2\beta) + \cdots$

$$C = \cos\alpha + n\cos(\alpha+\beta) + \frac{n(n-1)}{1\cdot 2}\cos(\alpha+2\beta) + \cdots$$

$$C+iS = (\cos\alpha + i\sin\alpha) + n\left(\cos(\alpha+\beta) + i\sin(\alpha+\beta)\right) + \frac{n(n-1)}{1\cdot 2}$$

$$\left[\cos(\alpha+2\beta) + i\sin(\alpha+2\beta)\right] + \cdots$$

$$= e^{i\alpha} + n\, e^{i(\alpha+\beta)} + \frac{n(n-1)}{1\cdot 2} e^{i(\alpha+2\beta)} + \cdots$$

$$= e^{i\alpha}\left[1 + n\cdot e^{i\beta} + \frac{n(n-1)}{2!} e^{i2\beta} + \cdots\right]$$

$$= e^{i\alpha}\left(1+e^{i\beta}\right)^n$$

$$= e^{i\alpha}\left(1+\cos\beta + i\sin\beta\right)^n$$

$$= e^{i\alpha}\left(2\cos^2\beta/2 + i\,2\sin\beta/2\cdot\cos\beta/2\right)^n$$

$$= e^{i\alpha}\left(2\cos\beta/2\right)^n\left(\cos\beta/2 + i\sin\beta/2\right)^n$$

$$= e^{i\alpha}\left(2\cos\beta/2\right)^n e^{i\beta/2\cdot n}$$

$$= \left(2\cos\beta/2\right)^n e^{i(\alpha+n\beta/2)}$$

$$C+iS = \left(2\cos\beta/2\right)^n\left(\cos(\alpha+n\beta/2) + i\sin(\alpha+n\beta/2)\right]$$

$$\Rightarrow S = \left(2\cos\beta/2\right)^n \sin(\alpha+n\beta/2)$$